Road-Network Trajectory Recovery via Constrained LLM Reasoning and Difficulty-Aware Query Routing

Supplementary Material

Appendix A
Paired Bootstrap Uncertainty and Voxel Scaling

Variance Concentration, Trip-Length Stratification, and Voxel Floors. Complementing Section IV-B of the main paper, Table I and Fig. 1 reveal three cross-split patterns. First, all 18 paired ΔMacro F1 intervals are strictly positive, yet CI half-widths against Deterministic C2 (± 0.0078–0.0118) are half those against RETRIEVE (± 0.0131–0.0215) and one-fourth those against VeTrac (± 0.0246–0.0431): Stage 1 gating ($\gamma_{\text{inv}} = 0$) and G_5^{kin} abstention share identical outputs across 88%–89% of routine corridors, zeroing paired variance outside multi-approach cohorts while independent neural baselines make uncorrelated errors split-wide. Second, ΔMacro F1 exceeds ΔMicro F1 by 6.3–12.1 points over VeTrac (short side-street underfitting) but matches within 0.4 points over RETRIEVE, showing uniform gains across trip lengths. Third, relative Vol-MSE reduction scales from 4.4%–24.8% on sparse UrbanVehicle (bounded by unobserved signal stops across $< 2.5\%$ camera coverage) to 54.7%–69.3% on dense Carla as C2 eliminates co-clustering detours globally.

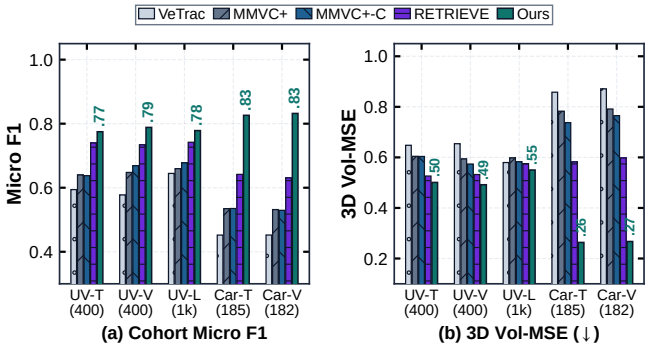


Fig. 1. Multi-split benchmark comparison complementing Fig. 5a of the main paper: (a) Cohort-Merged Micro F1 and (b) 3D Spatio-Temporal Volume MSE (Vol-MSE) across all five standard splits on UrbanVehicle and Carla.

Appendix B
Multi-Metric C2 Tiers and Seed Variance

Sensing-Dependent Tier Roles and Proximity Cushioning. Complementing Section IV-D of the main paper, Table II shows how camera density shifts tier workload: on sparse UrbanVehicle, Tier 1 interpolation (+20.7–22.0 points) and Tier 2 speed gating (+11.2–11.7 points) share recovery before C1–C3 resolves multi-approach turns, whereas on dense Carla, Tier 1 kinematic DP decoupling captures 98% of the F1 gain ($0.4732 \rightarrow 0.8069$) while Tier 3 zero-detour pruning adds $+2.14$ LCSS points and -5.0% Vol-MSE by removing opposite-lane snaps. Under outlier injection (Table III), continuous STLC degrades less than half as steeply as discrete F1 and LCSS under Tier 1+2 (-4.26 vs. -9.12 and -11.47 points at 20% noise) because retained anchors bound Euclidean proximity across severed spans until Tier 3's two-hop bypass ($v_{i-1}^* \rightarrow v_{i+1}^*$) heals link continuity and cuts seed variance threefold.

Appendix C
Cold-Start Scaling and Routing Frontier

Sublinear Bank Saturation and Slack-Insensitive Routing Plateau. Fig. 2 plots continuous cold-start and Pareto trajectories complementing Table VIII of the main paper. In Fig. 2a, marginal Macro F1 gain per indexed transition tapers sublinearly above 50% bank coverage ($+0.0482$ across $25\% \rightarrow 50\%$ vs. $+0.0267$ across $50\% \rightarrow 100\%$): once recurring arterial multi-approach junctions are populated, later experiences cover lower-frequency turns while G_5^{kin} keeps uncovered corridors at zero loss via abstention ($\mathcal{R}_q = \varnothing$). In Fig. 2b, relaxing routing slack from $\delta = 0.00$ to $\delta = 0.01$ triggers a $2.5\times$ step drop in latency ($4.84 \rightarrow 1.93$ s) followed by an invariant quality plateau across a $10\times$ slack span $\delta \in [0.01, 0.10]$: C1 dispatches $\mathcal{R}_q = \varnothing$ directly to GPT-4.1-nano, while Dual-GBR resolves cross-vendor near-ties within $\mathcal{R}_q \neq \varnothing$ toward faster GPT-4.1 at $\delta \in [0.01, 0.05]$ and gradually offloads low-branching $\mathcal{R}_q \neq \varnothing$ queries to GPT-4.1-nano as δ widens ($54.1\% \rightarrow 51.4\%$ strong tier at $\delta = 0.10$ on Val37; $56.2\% \rightarrow 52.1\%$ at $\delta = 0.05$ on Test48).

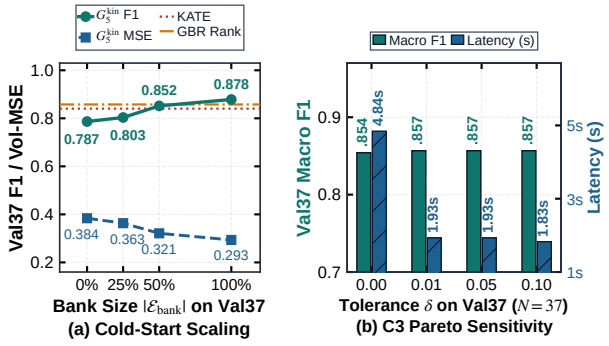


Fig. 2. Cold-start scaling and C3 Pareto sensitivity complementing Table VIII of the main paper: (a) Macro F1 and Vol-MSE on Val37 ($N = 37$) across $|\mathcal{E}_{\text{bank}}| \in \{0\%, 25\%, 50\%, 100\%\}$; (b) out-of-sample C3 quality–latency trade-off across $\delta \in \{0.00, 0.01, 0.05, 0.10\}$ on Val37.

TABLE I

PAIRED 5,000-RESAMPLE BOOTSTRAP 95% CONFIDENCE INTERVALS AND MULTI-METRIC GAINS OF **OURS (C1–C3)** OVER DOMAIN BASELINES AND ZERO-LLM PROGRAM CONTROLS ACROSS THE THREE HELD-OUT URBANVEHICLE SPLITS (COMPLEMENTING TABLE II OF THE MAIN PAPER).

Dataset Split	**Baseline / Program Control**	**Paired ΔMacro F1**	95% **Bootstrap CI**	**CI Half-Width**	**ΔMicro F1**	**ΔSTLC**
UrbanVehicle Test ($N = 400$)	Nearest-Node Only (0 LLM)	$+0.3465$	$[+0.3214, +0.3713]$	± 0.0249	$+0.3864$	$+0.1921$
	VeTrac	$+0.3014$	$[+0.2578, +0.3419]$	± 0.0420	$+0.1807$	$+0.1891$
	MMVC+	$+0.1490$	$[+0.1178, +0.1797]$	± 0.0309	$+0.1349$	$+0.0298$
	MMVC+-C	$+0.1498$	$[+0.1169, +0.1817]$	± 0.0324	$+0.1373$	$+0.0380$
	RETRIEVE	$+0.0351$	$[+0.0132, +0.0562]$	± 0.0215	$+0.0351$	$+0.0241$
	Deterministic C2 (0 LLM)	$+0.0257$	$[+0.0145, +0.0381]$	± 0.0118	$+0.0249$	$+0.0087$
UrbanVehicle Val ($N = 400$)	Nearest-Node Only (0 LLM)	$+0.3553$	$[+0.3306, +0.3787]$	± 0.0240	$+0.4006$	$+0.1970$
	VeTrac	$+0.3313$	$[+0.2878, +0.3741]$	± 0.0431	$+0.2109$	$+0.2101$
	MMVC+	$+0.1695$	$[+0.1395, +0.2008]$	± 0.0307	$+0.1406$	$+0.0426$
	MMVC+-C	$+0.1339$	$[+0.1050, +0.1632]$	± 0.0291	$+0.1196$	$+0.0254$
	RETRIEVE	$+0.0550$	$[+0.0345, +0.0752]$	± 0.0203	$+0.0544$	$+0.0358$
	Deterministic C2 (0 LLM)	$+0.0177$	$[+0.0081, +0.0292]$	± 0.0105	$+0.0169$	$+0.0056$
UrbanVehicle Large ($N = 1{,}000$)	Nearest-Node Only (0 LLM)	$+0.3880$	$[+0.3730, +0.4035]$	± 0.0152	$+0.4317$	$+0.2033$
	VeTrac	$+0.1969$	$[+0.1722, +0.2214]$	± 0.0246	$+0.1338$	$+0.1095$
	MMVC+	$+0.1342$	$[+0.1143, +0.1531]$	± 0.0194	$+0.1189$	$+0.0346$
	MMVC+-C	$+0.1032$	$[+0.0846, +0.1219]$	± 0.0186	$+0.1007$	$+0.0128$
	RETRIEVE	$+0.0323$	$[+0.0192, +0.0454]$	± 0.0131	$+0.0365$	$+0.0229$
	Deterministic C2 (0 LLM)	$+0.0217$	$[+0.0141, +0.0297]$	± 0.0078	$+0.0245$	$+0.0110$

TABLE II

COMPLETE FIVE-METRIC PROGRESSION OF C2 CORRIDOR VERIFICATION TIERS ACROSS SPARSE URBANVEHICLE VAL/TEST ($N = 400$) AND DENSE COLD-START CARLA TEST ($N = 185$, $\mathcal{E}_{\text{bank}} = \varnothing$), COMPLEMENTING TABLE IV AND SINGLE-METRIC TABLE V OF THE MAIN PAPER.

Dataset Split (N)	**Verification Tier / Recovery Configuration**	**Macro F1** ↑	**Micro F1** ↑	**STLC** ↑	**LCSS** ↑	**Vol-MSE** ↓
UrbanVehicle Val (400)	w/o C2 (Nearest-Node Anchors Only)	0.4133	0.3878	0.6938	0.3155	0.6638
	C2 Tier 1 Only (Unconstrained Shortest-Path)	0.6332	0.6912	0.8297	0.6356	0.5037
	C2 Tier 1+2 (+ Kinematic Span & Speed Gate)	0.7498	0.7693	0.8844	0.7474	0.5003
	C2 Tier 1–3 (Deterministic C2, 0 LLM Calls)	0.7509	0.7715	0.8852	0.7489	0.4997
	Full (C1–C3)	**0.7686**	**0.7884**	**0.8908**	**0.7643**	**0.4919**
UrbanVehicle Test (400)	w/o C2 (Nearest-Node Anchors Only)	0.4050	0.3885	0.6872	0.3083	0.6614
	C2 Tier 1 Only (Unconstrained Shortest-Path)	0.6123	0.6676	0.8161	0.6155	0.5195
	C2 Tier 1+2 (+ Kinematic Span & Speed Gate)	0.7238	0.7473	0.8692	0.7230	0.5127
	C2 Tier 1–3 (Deterministic C2, 0 LLM Calls)	0.7258	0.7500	0.8706	0.7248	0.5105
	Full (C1–C3)	**0.7515**	**0.7749**	**0.8793**	**0.7490**	**0.5010**
Carla Test (185)	w/o C2 (Nearest-Node Anchors Only)	0.4732	0.4976	0.7740	0.5291	0.8462
	C2 Tier 1 (Kinematic DP Corridor Separation)	0.8069	0.8209	0.8820	0.7374	0.2803
	C2 Tier 1+2 (+ Long-Gap Corridor Stitching)	0.8107	0.8242	0.8845	0.7429	0.2776
	C2 Tier 1–3 (+ Mid-Block Zero-Detour Pruning)	**0.8139**	**0.8262**	**0.8919**	**0.7643**	**0.2636**

TABLE III

MULTI-SEED ROBUSTNESS (MEAN ± STD ACROSS 5 RANDOM SEEDS) UNDER SYNTHETIC CAMERA OUTLIER CONTAMINATION ($\rho_{\text{noise}} \in \{5\%, 10\%, 20\%\}$) ON URBANVEHICLE TEST ($N = 400$), COMPLEMENTING TABLE V OF THE MAIN PAPER.

Noise ρ_{noise}	**C2 Verification Variant**	**Macro F1** ↑	**Micro F1** ↑	**STLC** ↑	**LCSS** ↑	**ΔMacro F1 vs. Clean**
5% Noise (5 seeds)	Tier 1 Only	0.5982 ± 0.0033	0.6575 ± 0.0028	0.8078 ± 0.0017	0.5953 ± 0.0035	-1.41 pts
	Tier 1+2	0.6974 ± 0.0031	0.7239 ± 0.0026	0.8531 ± 0.0021	0.6899 ± 0.0036	-2.64 pts
	Tier 1–3 (Deterministic C2)	$\mathbf{0.7232 \pm 0.0008}$	$\mathbf{0.7475 \pm 0.0006}$	$\mathbf{0.8685 \pm 0.0005}$	$\mathbf{0.7220 \pm 0.0008}$	$\mathbf{-0.26}$ **pts**
10% Noise (5 seeds)	Tier 1 Only	0.5922 ± 0.0071	0.6529 ± 0.0029	0.8011 ± 0.0032	0.5832 ± 0.0067	-2.01 pts
	Tier 1+2	0.6759 ± 0.0042	0.7049 ± 0.0025	0.8384 ± 0.0021	0.6618 ± 0.0053	-4.79 pts
	Tier 1–3 (Deterministic C2)	$\mathbf{0.7185 \pm 0.0017}$	$\mathbf{0.7429 \pm 0.0014}$	$\mathbf{0.8652 \pm 0.0008}$	$\mathbf{0.7159 \pm 0.0019}$	$\mathbf{-0.73}$ **pts**
20% Noise (5 seeds)	Tier 1 Only	0.5681 ± 0.0080	0.6320 ± 0.0038	0.7856 ± 0.0055	0.5502 ± 0.0098	-4.42 pts
	Tier 1+2	0.6326 ± 0.0068	0.6650 ± 0.0058	0.8116 ± 0.0046	0.6083 ± 0.0091	-9.12 pts
	Tier 1–3 (Deterministic C2)	$\mathbf{0.7068 \pm 0.0025}$	$\mathbf{0.7309 \pm 0.0030}$	$\mathbf{0.8570 \pm 0.0018}$	$\mathbf{0.7014 \pm 0.0033}$	$\mathbf{-1.90}$ **pts**